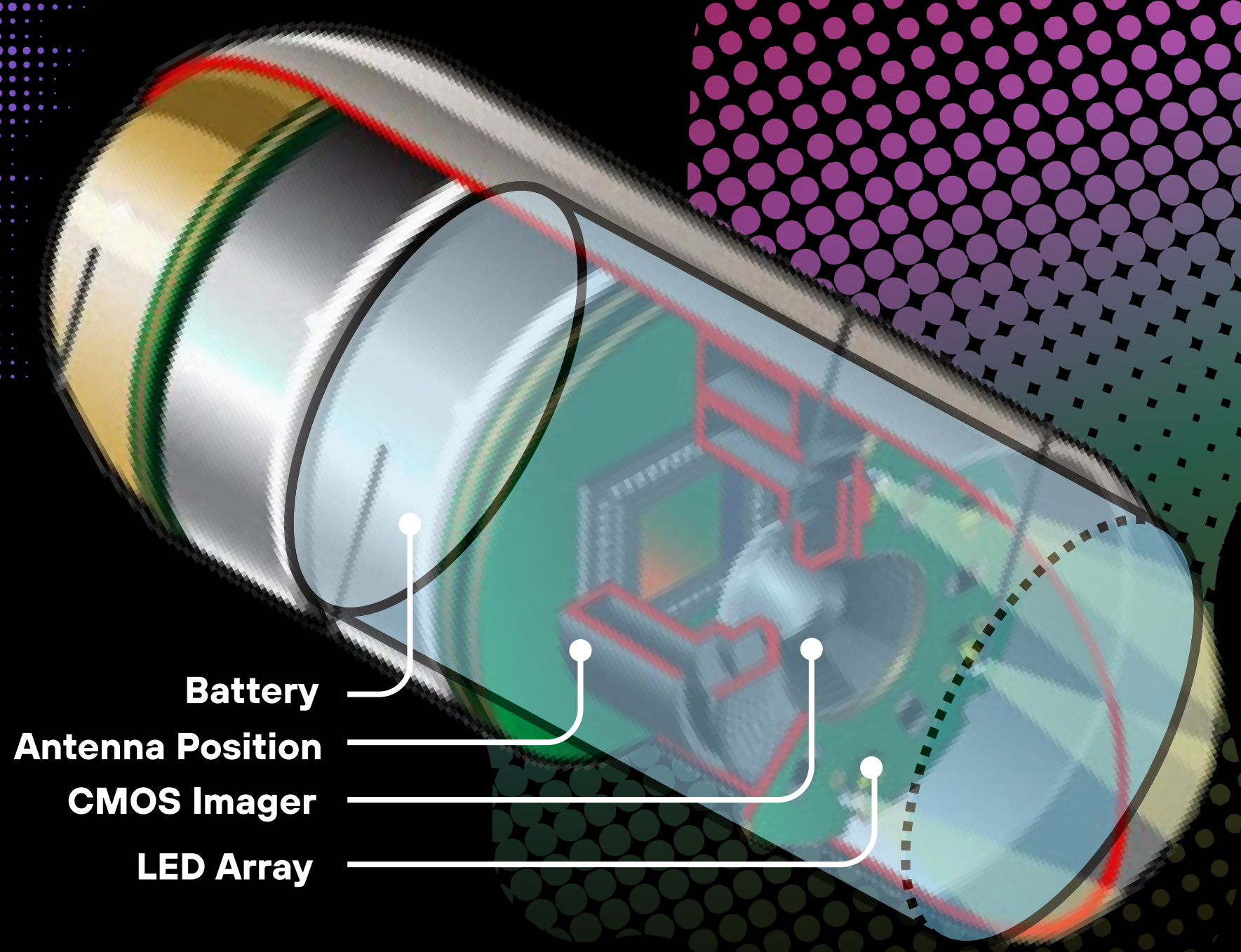


DOUBLE HELICAL

900MHZ ENDOSCOPIC PILL



Objective

To leverage 900MHz for capsule endoscopy, balancing between skin depth and data rate - potentially outperforming ~400MHz in data rate & 2.4GHz in tissue heating.

Dimension: 26mm (L) x 11mm (D)

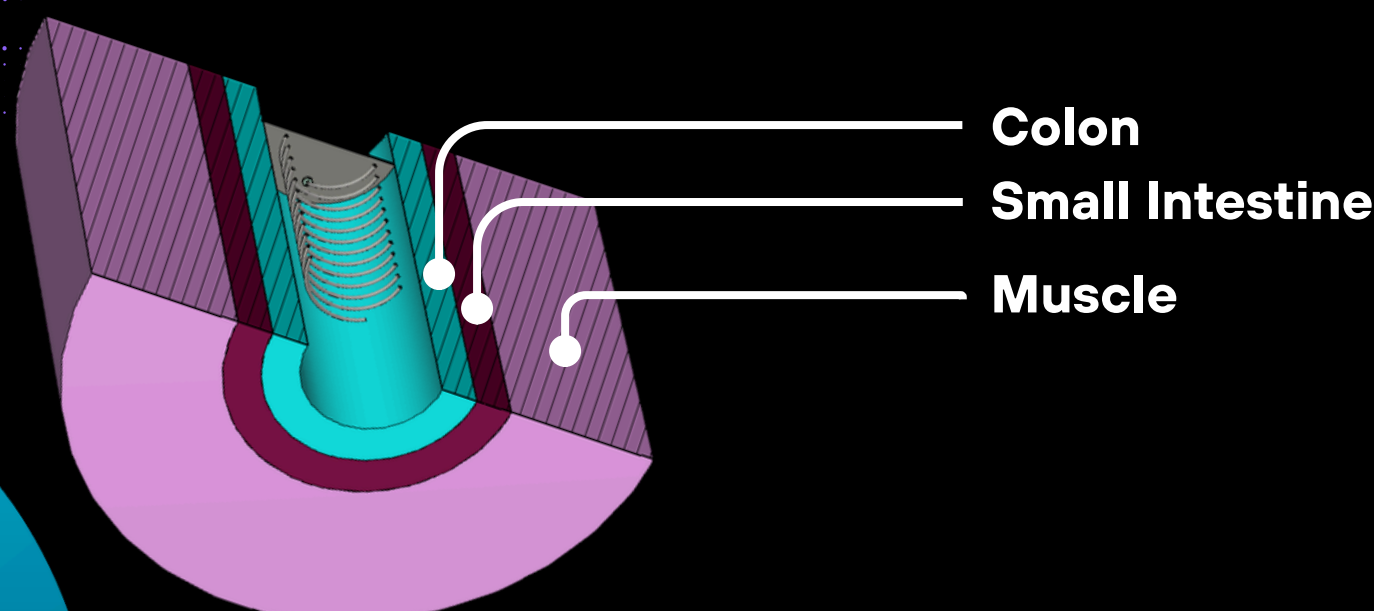
Design Considerations

Parameters	Current market Value	Simulated Value (w/ Phantom Model)
Size	$\leq 26\text{ mm} \times 11\text{ mm}$	22.31 x 7.8 mm
In-body Efficiency	~1%	~3%
Prioritized Parameter	Circular	Omni
Return Loss (dB)	$\leq -10\text{ dB}$	-21 dB
Skin Depth (mm)	11.4 mm	11.4 mm
SAR (maximum point)	2 W/Kg	0.5W/kg

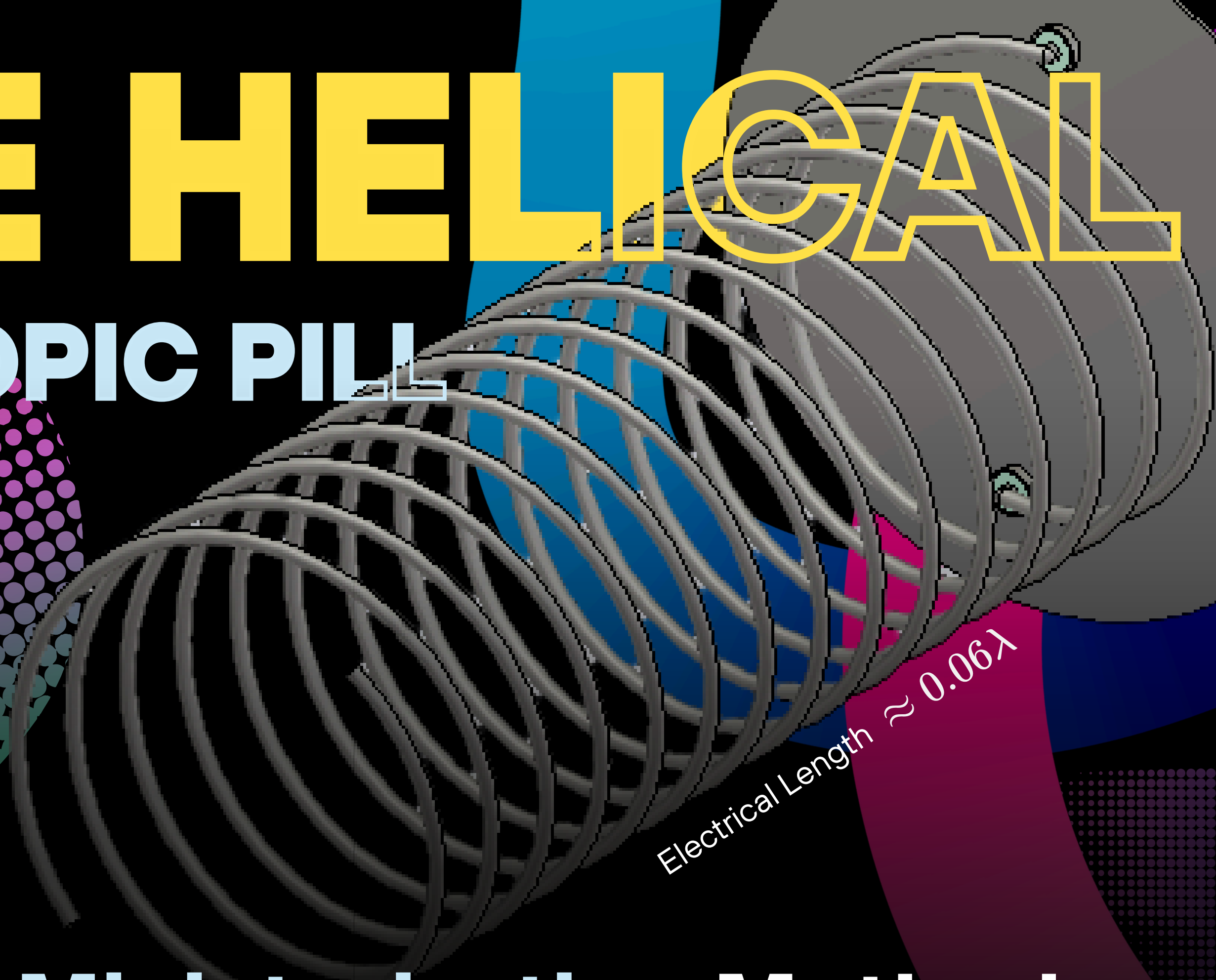
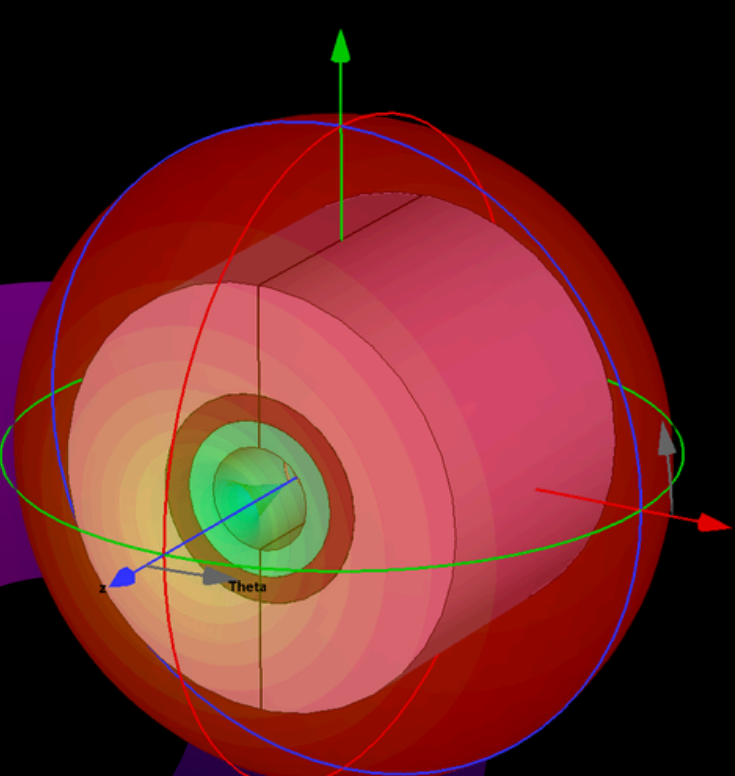
*SAR - Specific Absorption Rate (W/Kg)

Evaluation

Different tissue mediums were utilized to enable accurate evaluation of SAR, radiation pattern, and return loss, with results summarized in the table.



- ✓ SAR value achieved is below the safety limit of 2 W/kg
- ✓ Omnidirectional radiation pattern achieved. Ensures reliable signal coverage within human body.
- ✓ Good return loss (S11 parameter) Efficient power transfer and minimal signal reflection.



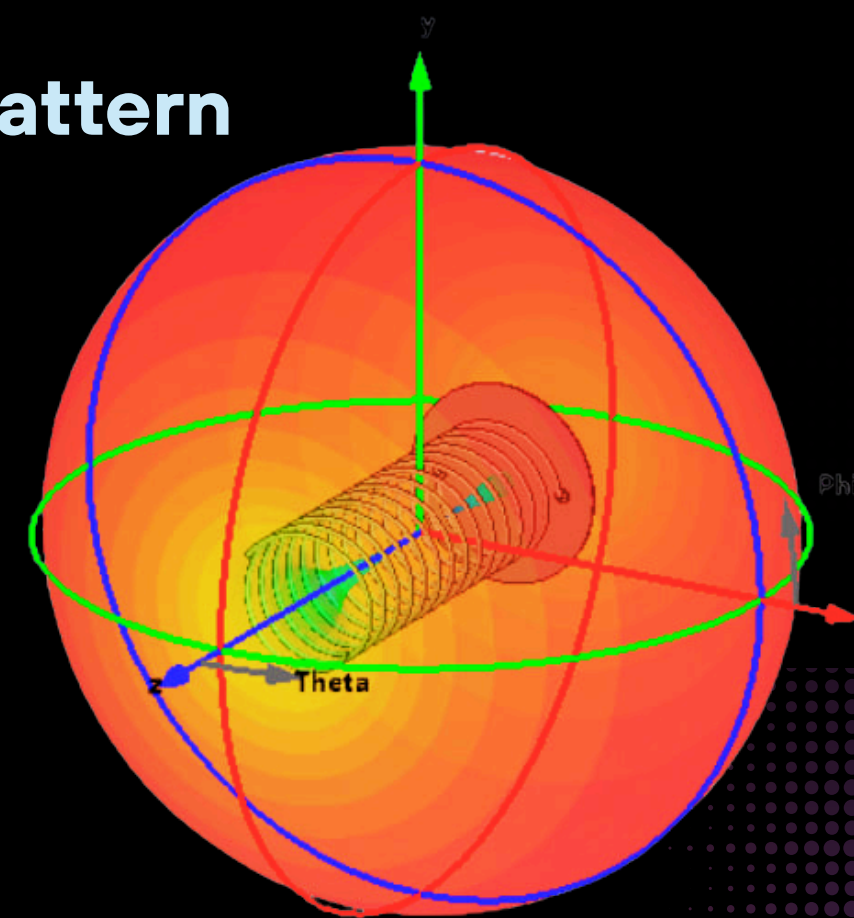
Electrical Length $\approx 0.06\lambda$

Miniaturization Method: DOUBLE FEED

Double feedback helical design was utilized to enhance the following key parameters:

- 1 Improved Efficiency & Radiation Pattern
- 2 Stable Circular Polarization

Gain	1.15dBi
Total Efficiency	92.5%
Return Loss (S11)	-8.1 dB (15.5% Reflected)



Phantom Model

The phantom model is a medium designed to mimic the electromagnetic properties of the human body. It allows for virtual simulations to be done, to ensure antenna performance—such as S₁₁, directivity, and tissue heating adhere to ICNIRP regulatory standards followed by Singapore.

- 1 Impedance Matching Optimization
- 2 Specific Absorption Rate Evaluation
- 3 Directivity & Radiation Patten Testing
- 4 Design Validation and Prototyping

Potential Explorations

900MHz Endoscopic pill would offer improved data rates & lesser tissue heating in theory, but practical challenges like body attenuation may impact real world performance. Further testing with the following components would serve for a fundamentally improved analysis of the antenna's design.

- 1 Physical Phantom Model
- 2 Endoscopic Pill Housing

